SPIRALS ON ONE PAGE

Spirals in parametric form

The same pair $(r\cos t,\ r\sin t)$ draws every spiral, and the gap between successive turns measures how tightly each one is wound.

1 / THE RULE AND THE PICTURE

A spiral is $(x, y) = (r(t)\cos t,\ r(t)\sin t)$, and only the function $r(t)$ changes from spiral to spiral



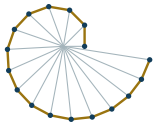
Archimedean
$r = at$



Fermat
$r = a\sqrt{t}$



logarithmic (golden)
$r = ae^{bt}$



Theodorus
$r_n = \sqrt{n}$

The Theodorus (square root) spiral stacks right triangles with a leg of 1, so its radii are the hypotenuses $\sqrt{2}, \sqrt{3}, \sqrt{4}, \ldots$; the first sixteen triangles are drawn, and the others show three turns of t.

2 / THE PARAMETRIC FORMS / ONE PARAMETER, TWO COORDINATES

Writing x and y separately in terms of t lets a curve wind round the origin as often as it likes

spiral	$r(t)$	$x(t)$	$y(t)$
Archimedean	at	$at\cos t$	$at\sin t$
Fermat	$a\sqrt{t}$	$a\sqrt{t}\cos t$	$a\sqrt{t}\sin t$
logarithmic	ae^{bt}	$ae^{bt}\cos t$	$ae^{bt}\sin t$
Theodorus	$\sqrt{n}$ at $t_n = \sum_{k<n} \arctan\frac{1}{\sqrt{k}}$	$\operatorname{Re} z_n$	$\operatorname{Im} z_n$, with $z_{n+1} = z_n\left(1 + \frac{i}{\sqrt{n}}\right)$

No equation $y = f(x)$ describes these curves, since a vertical line meets them many times, but the parameter t records how far round we are. The Theodorus rule is a complex multiplication: a step of length 1 at a right angle, $i\,z_n/|z_n|$, multiplies z_n by $1 + i/\sqrt{n}$ and takes the modulus from $\sqrt{n}$ to $\sqrt{n+1}$.

3 / PLOTTING FROM THE PARAMETRIC FORM / FIVE POINTS OF EACH SPIRAL

Choose $t = 0, \frac{\pi}{2}, \pi, \frac{3\pi}{2}, 2\pi$ and evaluate $(r\cos t,\ r\sin t)$

	$t=0$	$t=\frac{\pi}{2}$	$t=\pi$	$t=\frac{3\pi}{2}$	$t=2\pi$
Archimedean, $r=\frac{t}{2}$	$(0,\ 0)$	$(0,\ 0.785)$	$(-1.571,\ 0)$	$(0,\ -2.356)$	$(3.142,\ 0)$
Fermat, $r=\sqrt{t}$	$(0,\ 0)$	$(0,\ 1.253)$	$(-1.772,\ 0)$	$(0,\ -2.171)$	$(2.507,\ 0)$
golden, $r=e^{bt}$	$(1,\ 0)$	$(0,\ \varphi)$	$(-\varphi^2,\ 0)$	$(0,\ -\varphi^3)$	$(\varphi^4,\ 0)$

The golden spiral lands on powers of $\varphi = 1.618$ at each quarter turn because $e^{b\pi/2} = \varphi$, while the Archimedean radii add a constant 0.785 and the Fermat radii add less each time.

4 / THREE MEASURES OF TIGHTNESS / GAP, RATIO AND PITCH

The gap $g = r(t+2\pi) - r(t)$, the ratio $\rho = r(t+2\pi)/r(t)$ and the pitch $\tan\psi = r'/r$

spiral	gap g	ratio ρ	pitch $\tan\psi$	verdict
Archimedean	$2\pi a$	$\to 1$	$\frac{1}{t} \to 0$	constant spacing
Fermat	$\approx \frac{\pi a}{\sqrt{t}} \to 0$	$\to 1$	$\frac{1}{2t} \to 0$	ever tighter packing
logarithmic	$r(e^{2\pi b} - 1)$ grows	$e^{2\pi b}$ constant	b constant	self-similar
Theodorus	$\to \pi$	$\to 1$	$\to 0$	Archimedean for large n

The golden spiral has $b = \frac{2}{\pi}\ln\varphi = 0.3063$ and $\rho = \varphi^4 = 6.854$ per turn. Since $\theta_n \approx 2\sqrt{n} - 2.158$ for the Theodorus spiral, $r \approx \frac{t}{2} + 1.08$, an Archimedean spiral with $a = \frac{1}{2}$ and gap π.

5 / THE NUMBERS / GAP BETWEEN TURN k AND TURN $k+1$

The Theodorus gaps settle on π, the Fermat gaps shrink and the golden gaps grow sevenfold per turn

gap between turns	$k=1$	$k=2$	$k=3$	$k=4$	$k=5$
Archimedean, $r=\theta/2$	3.142	3.142	3.142	3.142	3.142
Theodorus, $r_n=\sqrt{n}$	3.150	3.145	3.143	3.143	3.142
Fermat, $r=\sqrt{\theta}$	1.038	0.797	0.672	0.592	0.535
golden, $r=e^{b\theta}$	40.125	275	1 885	12 920	88 555

The Theodorus gaps are interpolated between vertices at $t = 2\pi k$, and its first turn nearly closes at $n = 17$, at 6.129 radians.

6 / WHICH TIGHTNESS IS THE RIGHT ONE / WHAT EACH SPIRAL MODELS

The measure to use depends on what the spiral is for

A record groove keeps the constant Archimedean gap $2\pi a$, a sunflower places seed n at radius $c\sqrt{n}$ and angle $n \times 137.5°$ (Fermat's spiral at the golden angle, every seed equally crowded), and a nautilus grows by a fixed ratio per turn, the logarithmic spiral, which looks the same at every scale.

ASIDE / WHY PARAMETRIC

The parameter t is the angle here, and arc length is another choice of parameter for the same curves.

ASIDE / CHECK IN PYTHON

Iterating `z *= 1+1j/sqrt(k)` for $k = 1..16$ gives $|z| = \sqrt{17} = 4.123$ and `angle(z)` $= 6.129$ radians.

Convention. t is the angle in radians, $b = \frac{2}{\pi}\ln\varphi$; gaps use $a = \frac{1}{2}$ (Archimedean) and $a = 1$ (Fermat, golden).

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